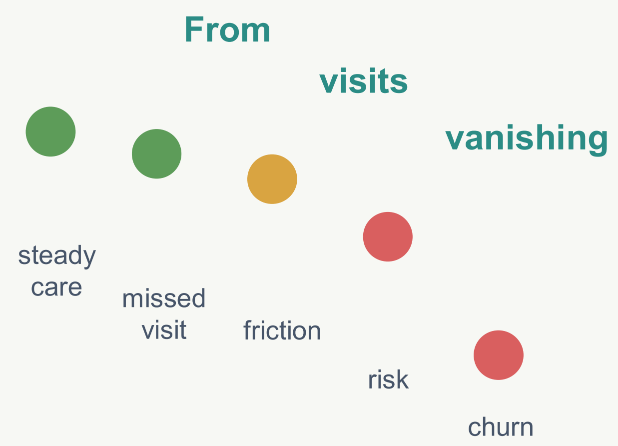
Why and When Do Patients Stop Coming Back to a Hospital?

A journey into disengagement in healthcare

Predicting who leaves is helpful.
Knowing when they begin to leave is powerful.

using **machine learning and survival analysis**.



DSCI 8750 • Data Science Capstone
Souleymane Diawara • Baba Sirima

The Silent Drop-Off - Motivation & Research Question

Patients do not leave overnight.
They disengage gradually until returning no longer feels worth it.

What disengagement looks like

- Missed appointments
- Longer gaps since the last visit
- Lower satisfaction with care
- Greater financial and access friction

Our goal

Detect early disengagement
Predict who is at risk
Understand when risk begins to accelerate

Keeps appointments
Stable care relationship

Misses a visit
Routine starts to weaken

Stops returning often
Distance, cost, or dissatisfaction grows

Quietly leaves care
The provider loses continuity and revenue

2,000 Patient Journeys

We tracked 2,000 patients across visits, missed appointments, satisfaction, financial burden, access, and portal engagement - then measured how long they remained active before churn.

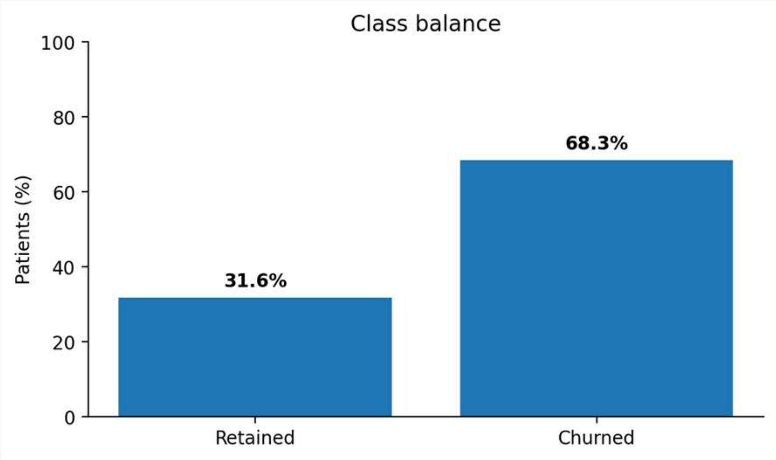
Because churn is a timeline, not a moment.

21
variables

0
missing values

0
duplicate rows

Tenure
time variable



The class balance is imbalanced: 68.35% churned and 31.65% retained.

✓ recall, ✓ sensitivity, and ✓ threshold choice matter

Demographics
•Age
•Gender
•State

Engagement behavior
•Visits_Last_Year
•Missed_Appointments
•Days_Since_Last_Visit
•Portal_Usage
•Referrals_Made

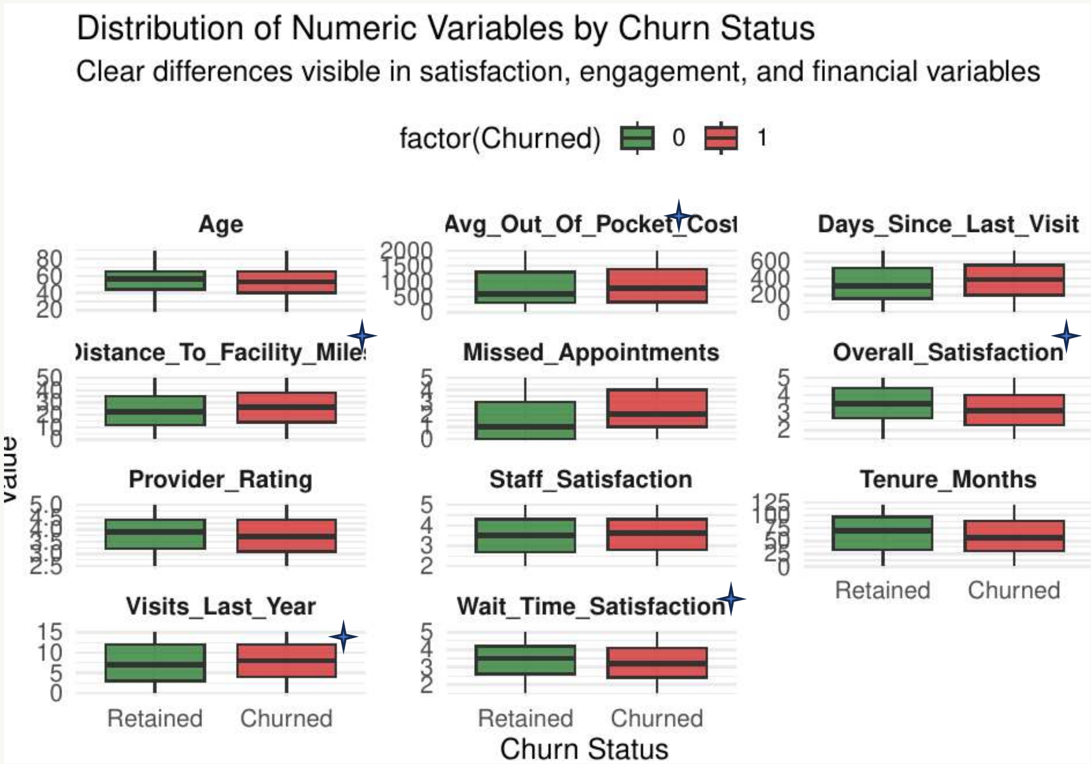
Satisfaction / experience
•Overall_Satisfaction
•Wait_Time_Satisfaction
•Staff_Satisfaction
•Provider_Rating

Access & financial friction
•Avg_Out_Of_Pocket_Cost
•Distance_To_Facility_Miles
•Billing_Issues

Time dimension
•Tenure_Months
•Last_Interaction_Date

Target
•Churned

Early Warning Signals

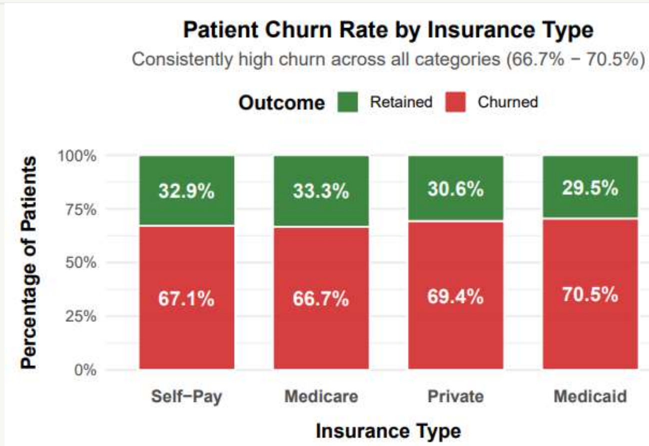


Clear differences appear in satisfaction, recency, cost, and access.

- Churned patients report lower overall and wait-time satisfaction.
- They have gone longer since their last visit and miss slightly more appointments.
- They also face higher out-of-pocket cost and live a little farther from the facility.

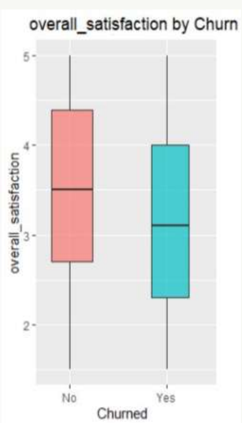
No single variable dominates. Disengagement seems to build through experience, behavior, and access together.

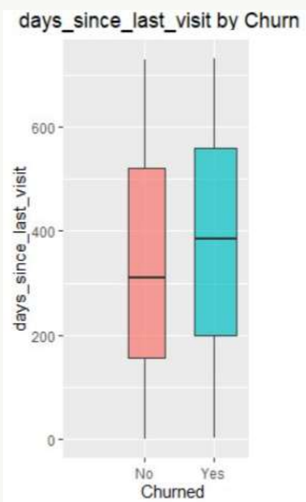
Access Matters, but Insurance Alone Does Not

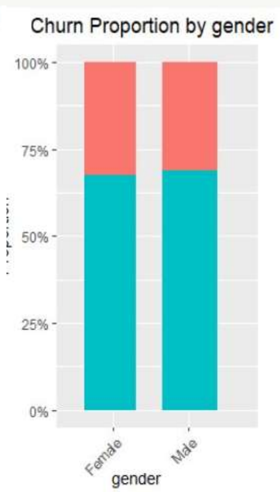


Churn stays high across every insurance type.

Medicaid is highest at 70.5%.
Medicare is lowest at 66.7%.
The spread is only 3.8 percentage points.





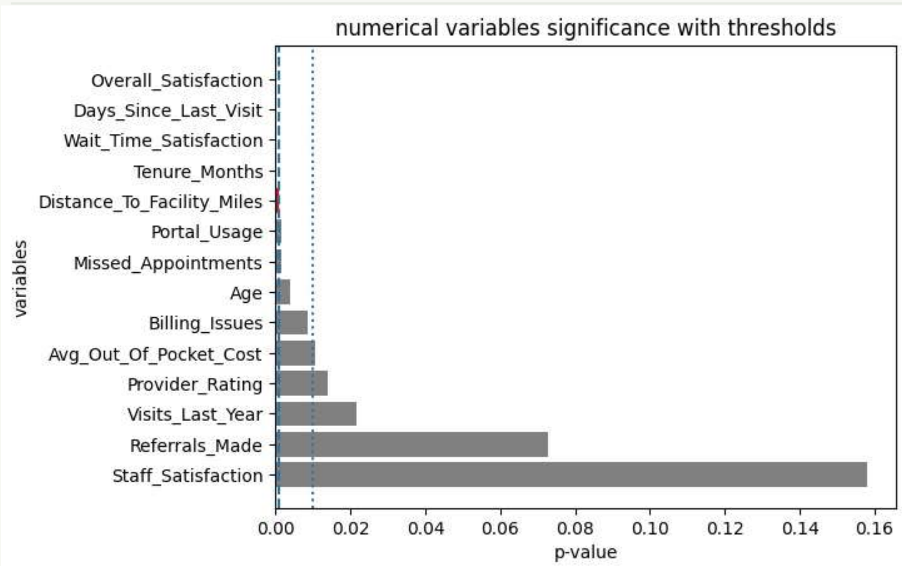


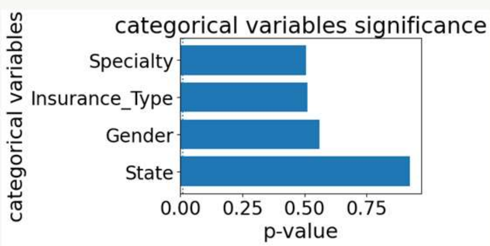
Interpretation

Insurance looks like a weak differentiator. System-wide experience and engagement factors appear stronger.

Chi-square tests in the supporting analysis showed categorical variables were not significant on their own.

Which Signals Matter Most?





Top statistical differences between retained and churned patients

Variable	Retained	Churned
Overall satisfaction	3.48	3.15
Days since last visit	339.1	378.6
Wait-time satisfaction	3.41	3.24
Tenure (months)	64.8	58.8
Distance (miles)	23.6	25.9

Cohen's d range ≈ 0.11 – 0.33

• Effects exist
• but each variable individually explains only a small part of churn
• explains why machine learning models perform better than single-variable tests.

Takeaway

The strongest signals are experience quality, visit recency, and access. Referrals made and staff satisfaction were not significant individually. *t-test / chi-test*

Predicting Risk Isn't Enough

We tested four "risk detectors" - but judged them by how well they separate churned from retained patients across thresholds.

Logistic Regression
Simple and interpretable
Best overall ROC AUC

Boosted Tree
Performance-focused
Captures nonlinear patterns

Random Forest
Strong specificity
But harder to explain

k-Nearest Neighbors
Similarity-based
Sensitive to local patterns

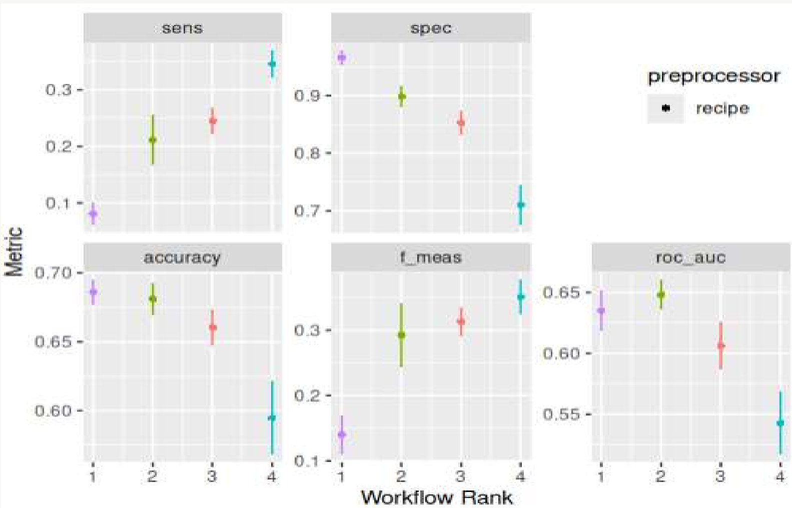
Evaluation design 80/20 train-test split • 5-fold cross-validation • Accuracy, sensitivity, specificity, F1, and ROC AUC

Why ROC AUC mattered most It tells us how well a model ranks future churners above non-churners across all possible cutoffs - exactly what early intervention needs.

Patient churn- sdi

Context · Signals · Prediction · Timing · Action

Which Risk Detector Worked Best?



Confusion Matrix [1]

(Random Forest)	Predicted Churn	Predicted Retained
Actual Churn	**48 (TP)**	**635 (FN)**
Actual Retained	**16 (FP)**	**301 (TN)**

(Log Regression)	Predicted Churn	Predicted Retained
Actual Churn	**143 (TP)**	**540 (FN)**
Actual Retained	**32 (FP)**	**285 (TN)**

Logistic Regression wins on ROC AUC.[2]

Model	Acc	Sens	Spec	F1	ROC AUC
Random Forest	.69	.07	.95	.14	.63
Logistic Reg.	.68	.21	.90	.29	.65
Boosted Tree	.66	.24	.85	.31	.60
k-NN	.59	.34	.71	.34	.55

Why not Random Forest?

It had slightly stronger accuracy and specificity, but its sensitivity (~0.07) was too low to reliably catch patients who were drifting toward churn.

Bottom line:
Logistic Regression offered the best balance of discrimination, stability, and interpretability.

Knowing when is much better : Cox Proportional Hazard

Prediction answers "who might leave?" Survival analysis answers "when does risk rise?"

Model time to event data
Assess how covariates affect risk over time

Function

$h(t|X)=h_0(t)\exp(\beta_1 X_1+\cdots+\beta_n X_n)$
Or
$\log\{h(t)\}=\log\{h_0(t)\}+\beta_1 X_1+\cdots+\beta_n X_n$

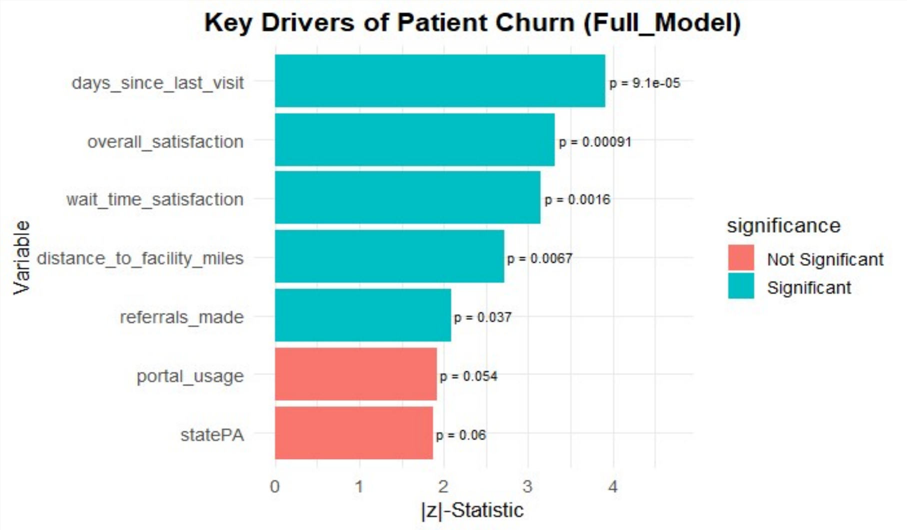
Components

$h(t)$: risk of leaving at time t
$h_0(t)$: baseline risk
$\exp(\beta)$: Hazard Ratio (HR)

Key Assumption

Proportional hazard
$h_1(t)/h_2(t)$ = constant over time

Which One Works Best



Model	Description	AIC ↓	Concordance ↑
Full model	All variables	18194.01	0.572
Reduced model	Only significant variables	18170.73	0.552

AIC (Akaike Information Criterion) measures model quality by balancing goodness-of-fit and complexity,

Lower AIC means better fit while avoiding unnecessary overfitting.

Proportional hazards assumption

GLOBAL p-value = 0.322 (> 0.05), so the assumption was satisfied. That means relative risk stayed stable enough over time for the Cox model to be valid.

This is the point where the project stops being only a churn classifier and becomes a timing model for patient disengagement.

What Changes the Timing of Churn?

Variable	β	HR	Translation
Overall satisfaction	-0.096	0.91	About 9% lower risk
Days since last visit	0.00049	1.0005	Risk rises 0.05% per day
Wait-time satisfaction	-0.090	0.91	About 8.6% lower risk
Referrals made	-0.05	0.95	About 5% lower risk
Distance to facility	0.0054	1.005	Risk rises 0.5% per mile

Interpretation Rule

HR = expβ
I If HR < 1 → ↓ Risk = (1−HR)×100
If HR > 1 → ↑ Risk = (HR−1)×100

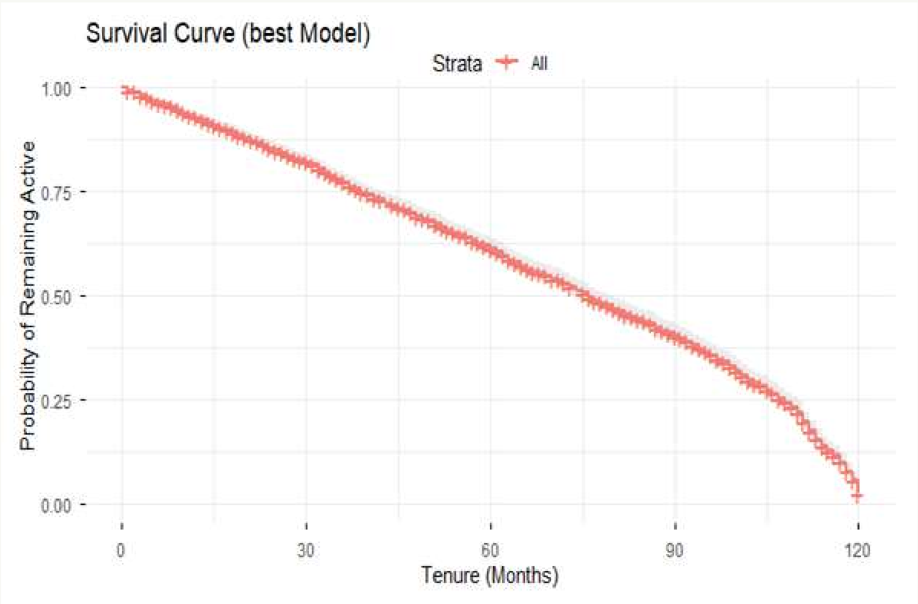
Higher satisfaction slows churn.

Longer gaps since the last visit quietly push risk upward.

Distance adds friction, which slowly increases time-to-churn risk.

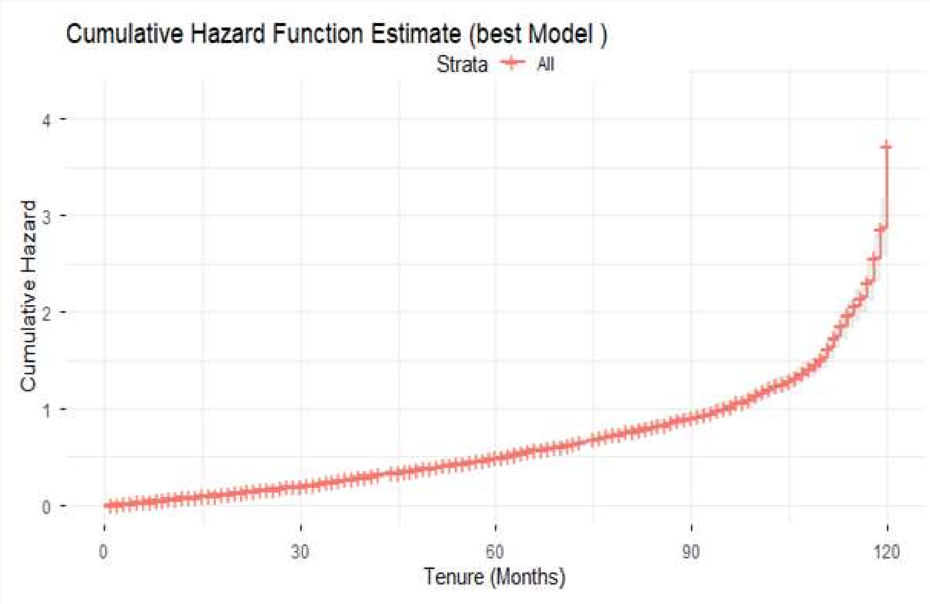
$$\text{Logh}\{(t)\} = \log\{h0\ (t)\} - 0.0962(\text{overall\_satisfaction}) + 0.00049\ (\text{days\_since\_last\_visit}) - 0.090\ (\text{wait\_time\_satisfaction}) - 0.05(\text{referrals\_made}) + 0.0054\ (\text{distance\_to\_facility\_miles})$$

Risk Builds Slowly, Then Accelerates



Probability of remaining active

The survival curve declines steadily and drops more sharply after about 90 months.



Cumulative hazard of churn

The cumulative hazard curve bends upward late in the timeline, showing acceleration in long-term risk.

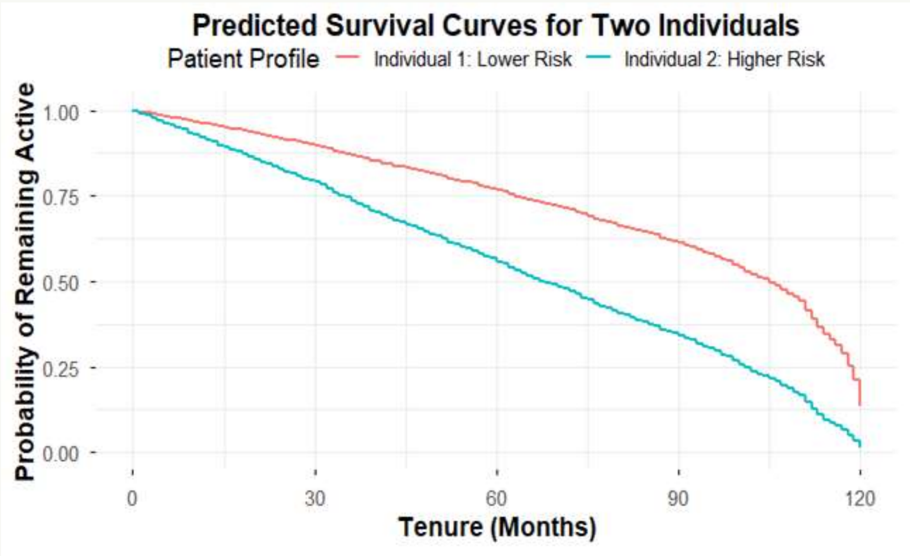
Same Clinic. Different Journeys.

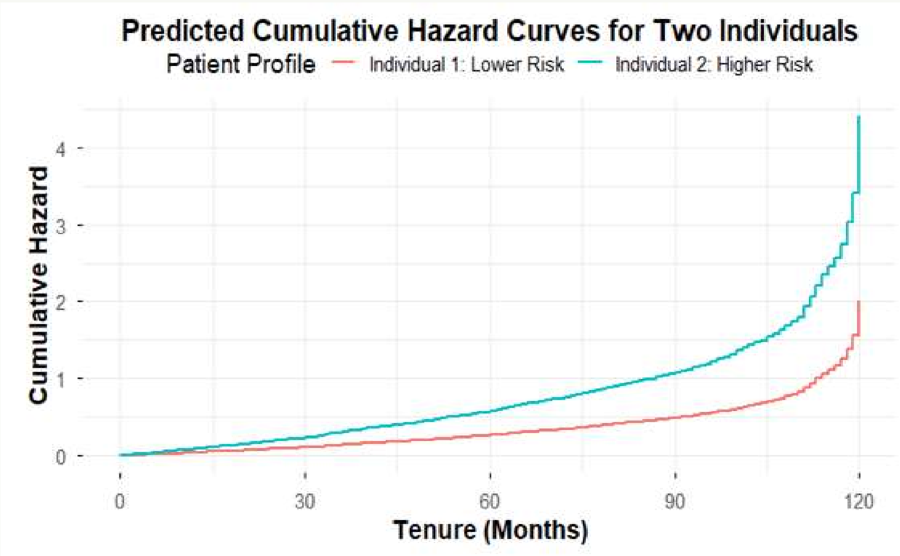
Lower-risk patient

High overall satisfaction • Recent visit (10 days)
Short distance (5 miles) • High wait-time satisfaction
Active engagement (2 referrals)

Higher-risk patient

Low overall satisfaction • Long time since visit (90 days)
Far distance (25 miles) • Low wait-time satisfaction
No engagement (0 referrals)





At month 60, survival is 76.7% for the lower-risk patient versus 55.9% for the higher-risk patient.

At month 75, cumulative risk is 0.37 versus 0.81 — same environment, very different outcomes.

What Drives Patients Away?

Lower satisfaction with care
Overall and wait-time satisfaction were among the strongest statistical signals.

Longer gaps since last visit
Days since last visit mattered in both the classification and survival analyses.

Access and cost friction
Distance and out-of-pocket cost both moved in the direction of higher churn risk.

Reduced engagement
Missed appointments and weaker digital/relationship touchpoints signaled drift.

**Machine learning helped identify who may leave.
Survival analysis revealed when risk begins to accelerate.**

Limitations

- Predictive performance is moderate rather than perfect.
- Cox PH assumes effects stay fairly constant over time.
- No single variable explains churn on its own.

This means patient retention is a human experience problem as much as a modeling problem.

Intervene Before Disengagement Becomes Departure

Low-risk patients
Maintain momentum

- Reinforce satisfaction and service quality
- Encourage routine visits and referrals
- Keep digital engagement active
- Protect continuity before friction appears

High-risk patients
Slow disengagement

- Trigger outreach when inactivity grows
- Reduce time since last visit
- Improve wait time and service experience
- Address distance and financial barriers early

Scientific recommendation

Use risk stratification and time-sensitive follow-up so intervention happens before the final visit becomes permanent churn.

Final takeaway

Patients do not suddenly leave care
They slowly disengage until returning no longer seems worth it.


After waiting too long...
SORRY FOR THE WAIT!
BARBER SHOP
I'M NEVER COMING BACK!
That's Churn.
SORRY I AM LATE FOR MY CAPSTONE AT UNO

Q & A